

M3Grounder: Mask-Based Multi-Span and Multi-Granular Grounding for Document QA

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Abstract

*Document QA requires not only accurate answers but also identifying where each answer is grounded on the page. Most approaches treat the task as text-only generation, while existing answer grounding methods generate coarse bounding boxes that fail to capture curved text. We introduce **M3Grounder**, a hybrid vision-language and segmentation architecture that formulates document grounding as pixel-level segmentation. It produces fine-grained evidence masks refined by a bleed-suppression loss to prevent spillover. M3Grounder autoregressively generates answer text interleaved with [GROUND] tokens that link individual answer spans to their corresponding evidence regions. Also, M3Grounder grounds evidence hierarchically across phrase, line, and block levels using an enclosure loss that enforces spatial containment. We release **GroundingDocQA** dataset (200K documents, 2M multi-span and multi-granular QA pairs with pixel-level grounding masks), built through a data engine that handles complex layouts, curved-text and graphic-rich documents. We also release **GroundingDocQA-Bench**, a diverse and challenging human-verified benchmark. M3Grounder sets a new state-of-the-art in grounded DocVQA, advancing from coarse boxes to hierarchical, fine-grained and contextually grounded mask evidence. Code, dataset and models will be released.*

1. Introduction

Document question answering requires Vision-Language Models (VLMs) [3, 4, 10, 19, 35, 50, 53] to reason jointly over text and spatial layout. Most approaches [15, 16, 30] treat DocVQA as pure text generation and ignore evidence localization. This weakens reliability, especially in domains such as healthcare, law, and finance where

grounding is crucial. Existing grounding methods [50, 53, 65] generate bounding boxes that offer only coarse localization. Rectangular regions are not suitable for curved text and often capture irrelevant background, leading to ambiguous grounding.

To address these limitations, we introduce **M3Grounder**, a novel hybrid VLM-Segmentation architecture that models document QA grounding as pixel-level segmentation instead of box prediction. M3Grounder generates answers interleaved with special [GROUND] tokens, each linked to a segmentation mask representing a distinct evidence region. Unlike prior grounding methods [50, 53, 65] that interleave bounding box coordinates within the textual answer, M3Grounder decouples language modeling from spatial grounding prediction. This separation allows the VLM to focus on producing semantically accurate answers and span boundaries, while the segmentation head handles fine-grained, geometry-aware localization.

Consequently, M3Grounder produces precise masks that align with text regions, preserving shape and curvature (Fig. 1A,B). A bleed-suppression loss refines localization by penalizing spillover into irrelevant areas. Also, our approach enables multi-span grounding, allowing answers to draw evidence from multiple, spatially distinct regions of a document (Fig. 1B).

Documents follow a natural spatial hierarchy: words form lines, and lines form blocks, each granularity capturing progressively broader layout structure. These granularities support different reasoning scopes in QA, providing fine-grained spans for extractive questions (e.g., “Name:” or “Address:”) and broader regions for abstractive ones (e.g., “Summarize the reason for acceptance”). As grounding moves from phrases to lines and blocks, the added context captures neighboring cues, enhancing interpretability. Motivated by these observations, M3Grounder grounds evidence across phrase, line, and

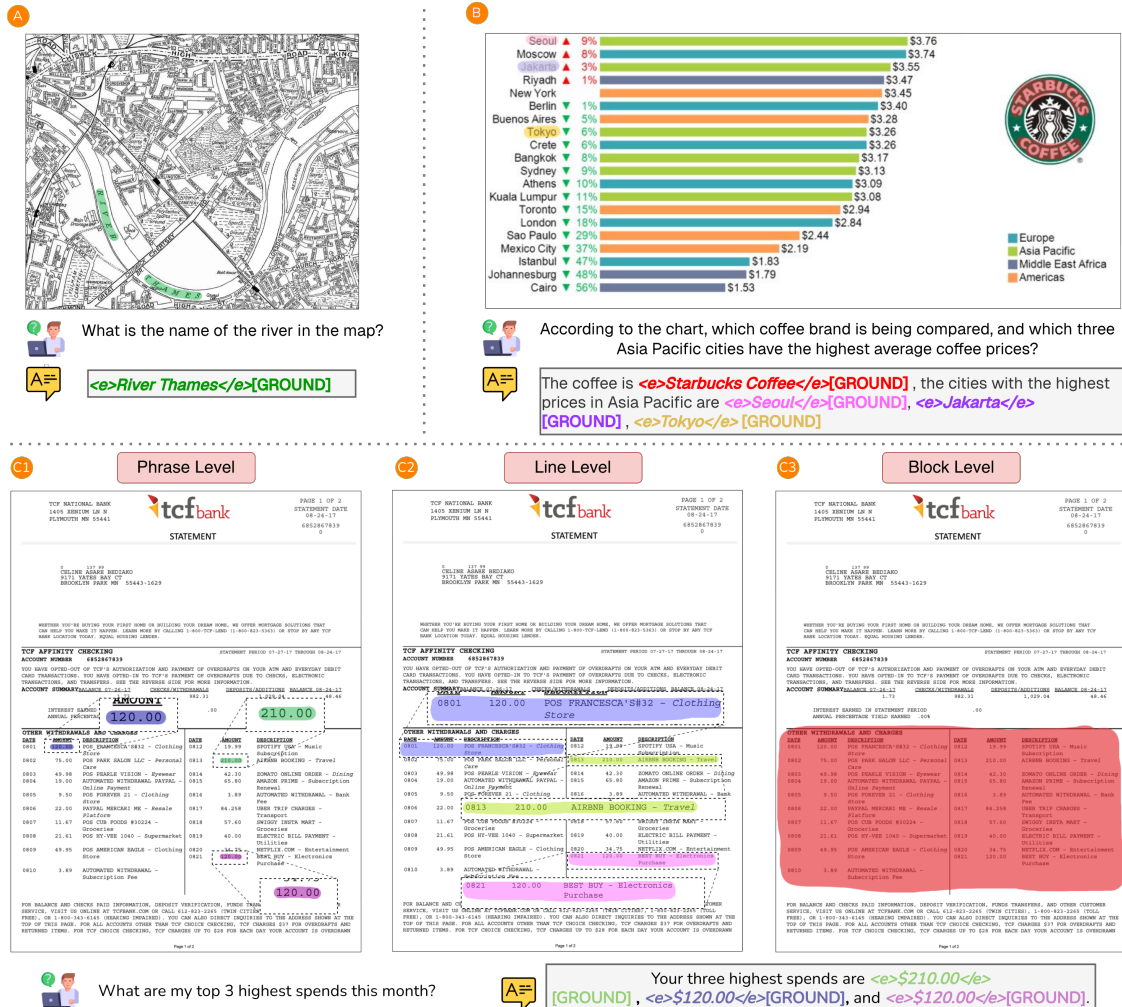


Figure 1. **M3Grounder in action:** Each example shows a QA pair. The predicted answer text contains interleaved [GROUND] tokens which map answer spans to corresponding grounding regions. **A**, **B** demonstrates precise segmentation without spillover into irrelevant regions. **B** shows effective grounding for dense, multi-span evidence in complex document layouts. **C1**, **C2**, and **C3** illustrate multi-granular grounding, where the grounding scope expands hierarchically (phrase $\subset$ line $\subset$ block).

block levels (Fig. 1C). A *hierarchical enclosure loss* enforces nested containment (phrase $\subset$ line $\subset$ block) of grounding regions, ensuring spatial consistency. Empirically, this optimization also improves grounding precision and QA accuracy.

Existing document QA grounding datasets [14, 65] are predominantly limited to extractive answers and contain grounding only in terms of bounding boxes. They also lack the hierarchical cues required for multi-granular QA grounding. To enable fine-grained and hierarchical supervision, we develop a novel data engine that generates *mask-based*, *multi-span* and *multi-granular* grounded QA pairs across heterogeneous document types. We use this data engine to construct **GroundingDocQA**, a large-scale dataset of 200K documents and 2M QA pairs, containing grounding regions with varying curvature.

For evaluation, we introduce **GroundingDocQA-Bench**, a human-verified benchmark of 2.5K documents and 5K QA pairs to jointly assess answer accuracy and grounding fidelity under multi-span, multi-granular, and curved-text settings. Across diverse benchmarks, M3Grounder improves grounding accuracy, raising benchmark scores by large margins and establishing a new state of the art in grounded document QA.

2. Related Work

Grounding has become a key capability for enhancing interpretability in multimodal models [7, 22, 38, 41, 58, 61, 62]. While these models align textual content with visual entities, they are designed for natural scenes and transfer poorly to documents [65].

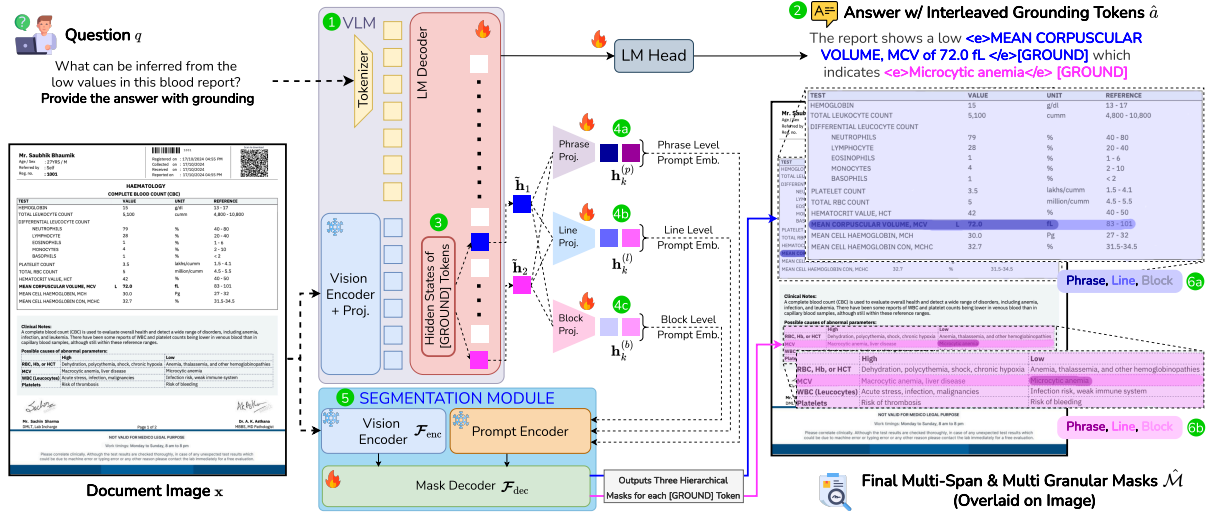


Figure 2. **Overview of M3Grounder:** Given a document image and question, a VLM ① encodes visual features and autoregressively generates an answer sequence ② interleaved with [GROUND] tokens. Hidden states of the [GROUND] tokens ③ $\tilde{h}_1, \tilde{h}_2$ from the VLM decoder are passed through granularity-specific MLP projection heads, producing phrase, line, and block level prompt embeddings $\mathbf{h}_k^{(p)}, \mathbf{h}_k^{(l)}, \mathbf{h}_k^{(b)}$ ④. In parallel, the promptable segmentation module ⑤ extracts dense image embeddings reused across all spans and granularities. The prompt embeddings ④ are decoded by the mask decoder $\mathcal{F}_{dec}$ to generate hierarchical grounding masks ⑥, mapping each answer span to its corresponding grounding masks across multiple granularities.

Recent document understanding models [50, 53, 65] support question answering with grounding. General-purpose VLMs such as Qwen3-VL [50] and InternVL3.5 [53] output bounding boxes for evidence localization but are not explicitly optimized for grounding accuracy. DOGR [65] jointly predicts answers and bounding boxes, improving over general-purpose VLMs but remains limited to coarse rectangular localization at a single granularity. In contrast, **M3Grounder** provides *mask-based hierarchical grounding* that captures phrase-, line-, and block-level structure, enabling more precise and interpretable evidence alignment.

Many existing document QA datasets [31–33] provide QA pairs without grounding. Recent efforts such as DOGR [65] and BoundingDocs [14] rely on box-level grounding derived from OCR spans, and predominantly restrict evaluation to extractive settings. Our **GroundingDocQA** and **GroundingDocQA-Bench** advance this landscape by introducing *fine-grained, multi-granular, mask-based annotations* that capture hierarchical document structure and enable comprehensive evaluation of grounding under diverse, real-world layouts.

3. Method

Given a document image x and question q , **M3Grounder** autoregressively generates an answer $\hat{a} = (\dots \langle e \rangle y_k \langle /e \rangle [\text{GROUND}] \dots)$ (see Fig. 2). Each [GROUND] token triggers generation of hierarchi-

cal masks $\{\hat{\mathbf{M}}_k^{(i)} \mid i \in \mathcal{G}\}$ for granularities $\mathcal{G} = \{p, l, b\}$ (phrase, line, block). Formally, the model f outputs a joint prediction $f_\theta(x, q) \rightarrow (\hat{a}, \hat{\mathcal{M}} = \{\hat{\mathbf{M}}_k^{(i)}\}_{k=1, i \in \mathcal{G}}^K)$, where K is the number of grounded spans (see Fig. 2). If $K = 0$, we define $\hat{\mathcal{M}} = \emptyset$ (i.e., no grounding masks are produced).

3.1. Model Architecture

M3Grounder couples a Vision–Language Model with a promptable segmentation module. The VLM encodes (x, q) and autoregressively generates the answer interleaved with [GROUND] tokens¹. Each [GROUND] token is emitted immediately after its answer span, establishing a one-to-one span-region link for precise grounding. The token’s final hidden state $\tilde{h}_k$ is projected by three MLP heads to produce granularity-specific prompts $\mathbf{h}_k^{(i)}$ for $i \in \{p, l, b\}$ (phrase, line, block). In parallel, the segmentation module extracts dense visual features $\mathbf{z} = \mathcal{F}_{enc}(x)$. These features are decoded with $\mathbf{h}_k^{(i)}$ via $\mathcal{F}_{dec}$ to produce grounding masks $\hat{\mathbf{M}}_k^{(i)}$. Each mask thus captures phrase-, line-, or block-level evidence for its linked answer span, enabling *span-to-region grounding*.

3.2. Training Objectives

M3Grounder is trained end-to-end to jointly optimize text generation and mask prediction. The total loss combines language modeling, segmentation, and two auxil-

¹[GROUND] is a special token added to vocabulary.

inary terms that enhance spatial precision and hierarchical consistency:

$\mathcal{L}_{\text{total}} = \lambda_{\text{lm}}\mathcal{L}_{\text{lm}} + \lambda_{\text{seg}}\mathcal{L}_{\text{seg}} + \lambda_{\text{bleed}}\mathcal{L}_{\text{bleed}} + \lambda_{\text{hier}}\mathcal{L}_{\text{hier}}$ where $\mathcal{L}_{\text{lm}}$ is the standard cross-entropy loss [42] computed over all output tokens for accurate QA generation. Each predicted mask $\hat{\mathbf{M}}_k^{(i)}$ is supervised against its ground truth $\mathbf{M}_k^{(i)}$ using a combination of Dice [34] and binary cross-entropy losses [42]:

$$\mathcal{L}_{\text{seg}} = \sum_{i \in \mathcal{G}} \sum_{k=1}^K \left[\lambda_{\text{dice}}^{(i)} \ell_{\text{dice}}(\hat{\mathbf{M}}_k^{(i)}, \mathbf{M}_k^{(i)}) + \lambda_{\text{bce}}^{(i)} \ell_{\text{bce}}(\hat{\mathbf{M}}_k^{(i)}, \mathbf{M}_k^{(i)}) \right]$$

Bleed Suppression ($\mathcal{L}_{\text{bleed}}$): To prevent mask spillover into non-text regions, we introduce a bleed suppression term that penalizes overlap beyond text pixels. Let $\mathbf{M}_{\text{ref}}$ denote the union of all text regions and Ω represent all pixel coordinates. We have:

$$\mathcal{L}_{\text{bleed}} = \sum_{i \in \mathcal{G}} \sum_{k=1}^K \frac{\sum_{j \in \Omega} \hat{\mathbf{M}}_k^{(i)}(j) [1 - \mathbf{M}_{\text{ref}}(j)]}{\sum_{j \in \Omega} \hat{\mathbf{M}}_k^{(i)}(j) + \epsilon}$$

This loss encourages predicted masks to be confined strictly within valid text areas by penalizing any portion that extends into background regions.

Hierarchical Enclosure ($\mathcal{L}_{\text{hier}}$): To ensure finer masks remain within coarser ones, we define:

$$\mathcal{L}_{\text{hier}} = \sum_{k=1}^K \sum_{(i,j) \in \{(p,l), (l,b)\}} \frac{\sum_{m \in \Omega} \hat{\mathbf{M}}_k^{(i)}(m) [1 - \mathbf{M}_k^{(j)}(m)]}{\sum_{m \in \Omega} \hat{\mathbf{M}}_k^{(i)}(m) + \epsilon}$$

This enforces a containment hierarchy ($p \subset l \subset b$) and improves spatial alignment. Together, these objectives ensure accurate QA generation, precise mask localization, and consistent multi-granular grounding.

4. Dataset

We introduce **GroundingDocQA**, a large-scale dataset for mask-based, multi-span, and multi-granular document grounding. It contains 2M QA pairs across 200K documents, including charts, reports, forms, tables, webpages, and infographics—capturing diversity of real-world documents. The dataset is constructed through three complementary pipelines: ① 140K text-rich documents that retain standard layouts [12, 24, 32, 33, 47, 54], ② 10K documents with curved and skewed text [23, 28, 56, 63] to capture irregular geometry, and ③ 50K charts [21, 26, 59] to model structured visual reasoning. Each QA pair is annotated with pixel-level masks across phrase, line, and block granularities, enabling precise spatial grounding.

4.1. Data Generation Pipeline

Layout-Aware Documents: Prior document grounding datasets [14, 65] often generate QA pairs using VLMs and ground answers by matching the predicted text to OCR spans or bounding boxes. Such grounding is unreliable when OCR is noisy or the same phrase appears in multiple locations. Moreover, these pipelines flatten documents into unstructured text sequences, discarding layout hierarchy and losing the spatial context that is central to document understanding.

To overcome these issues, we convert each document into a high-fidelity, layout-aware HTML variant using the REPLICA engine [2] (Fig. 3 1A). This converted HTML form, referred to as *Fid-HTML*, preserves the document’s spatial structure, reading order, and the hierarchical organization of lines and blocks, providing a faithful representation for QA generation.

Each HTML element is assigned a unique ID, ensuring traceability across structure levels. The ID-tagged *Fid-HTML* is then provided as input to a LLM [1], which is prompted to generate QA pairs, the associated line-level HTML IDs and certain key phrases. To extend each QA pair 1B with multi-granular grounding, we first obtain lines by mapping the ID tags back to their bounding boxes. This ensures spatial consistency while avoiding hallucinated coordinates. Next, we obtain phrase-level boxes by using the key phrases in a post-processing step. Block-level boxes are obtained from the enclosing HTML hierarchy of each line (dashed blue/pink boxes in 1C). For table-related questions, the answer cell corresponds to the phrase-level box, the cell’s row to the line-level box, and the entire table to the block-level box. All boxes are mapped to pixel-level segmentation masks using a postprocessing module 1E [60]. This step preserves the geometric contours of text and produces precise phrase, line, and block levels 1G.

Curved Text Documents: Many real-world documents contain curved or irregular text layouts. Bounding boxes fail to precisely capture these layouts. Hence, we introduce a dedicated pipeline that generates QA pairs grounded on curved text regions. Given a document image, we first apply a curved text segmentation module 2A to obtain pixel-level curved masks. We overlay each mask and denote it as a highlight (e.g. irregular black dashed polygon in 2B). These highlights are used as prompts [57] by a VLM [4] to generate QA pairs 2C. This ensures each QA pair remains precisely aligned with its corresponding segmentation mask, enabling reliable supervision for curved text grounding.

Charts: Interpreting charts require understanding numerical patterns, relational trends, and comparative structures. To capture this complexity, we curate a chart dataset covering diverse chart types—bar, line, pie, scatter, stacked,

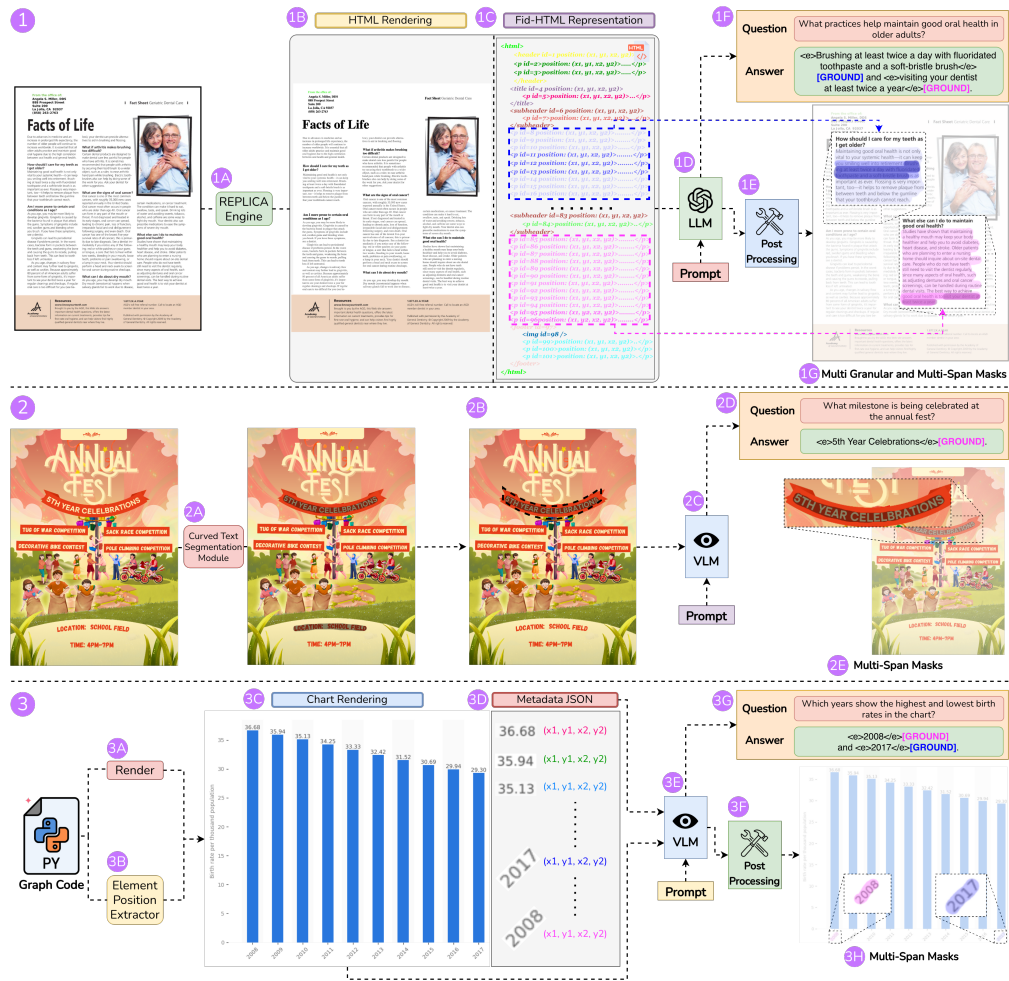


Figure 3. **GroundingDocQA generation pipeline:** The data engine generates grounded QA pairs from three pipelines. (1) Text-rich documents are converted into FID-HTML using the REPLICA engine [2] (1A). The HTML representation is input to a suitably prompted LLM (1D) and processed to obtain layout-aware QA generation with multi-granular segmentation masks (1G). (2) Curved and irregular text documents are processed via a segmentation module and curvature detection algorithm (2A) [5]; selected irregular masks are used to generate grounded QA pairs (2D). (3) Charts are rendered from plotting scripts, and element coordinates are captured directly during rendering (3B) to generate QA pairs grounded to chart components (3E). Together, these pipelines provide consistent pixel-level grounding across structured layouts, curved text, and visual charts.

mixed, and others across a wide range of visual styles. We utilize publicly available chart datasets [21, 26, 59] that provide executable scripts for rendering charts. Each script is executed to generate a chart image, during which we intercept the renderer functions of the plotting libraries [13, 18, 37, 55] (3B) to record the bounding boxes of all chart elements. This direct interception avoids detection errors, ensuring precise localization of chart components at render time.

For every chart element (axes, ticks, legends, titles), we assign a unique ID, extract its bounding box, textual content, visual type and store in a JSON file (3D). The rendered chart image, along with its ID-tagged JSON, is

provided as input to a VLM [4], which is prompted to generate QA pairs (3G) along with the corresponding element IDs. These IDs are then mapped back to their bounding boxes and converted into pixel-level segmentation masks during post-processing [60] (3H). Unlike text documents, charts have a single level grounding representation.

4.2. Data Verification Pipeline

We employ a two-stage verification pipeline that checks both semantic and factual consistency across all QA pairs. In the first stage, we verify semantic alignment between each answer span and its corresponding grounded region. For this, we compute the sentence-embedding similarity [52] between the span’s text and text obtained by

OCR-ing the region crop. QA pairs with low similarity scores indicate unreliable text-region alignment and therefore removed. In the second stage, we assess factual and logical consistency of text documents using a verifier model (LLM-as-a-Judge) [11, 64]. The verifier takes the document HTML, QA pair as inputs and evaluates the correctness of the answer. QA pairs that fail this support check are discarded. This stage ensures that the final dataset retains only factually grounded and logically coherent annotations.

4.3. GroundingDocQA-Bench

Existing document QA grounding benchmarks [14, 65, 67] contain only bounding box grounding. Therefore, we introduce **GroundingDocQA-Bench**, a segmentation-based benchmark. We manually select document images from private sources as well as from the test sets of various public document datasets [21, 23, 26, 28, 32, 33, 47, 56, 59, 63]. Documents with curved and skewed text are specifically included to enable robust geometric and structural evaluation. Apart from regular text-rich documents, we specifically include other popular document categories such as charts, reports, forms, tables, web-pages and infographics for improved diversity. We generate QA pairs using our data engine (Sec. 4). Each QA pair is annotated by both text geometry-straight (70%) or curved (30%) and span type (2,820 single-span and 2,180 multi-span instances). The benchmark is manually curated and verified to ensure quality, and unbiased evaluation. Overall, GroundingDocQA-Bench contains 2.5K documents and 5K QA pairs.

5. Experiments

Implementation Details: M3Grounder couples a VLM backbone with a promptable segmentation head. We explore two backbone variants: M3Grounder-I (InternVL3.5-8B) [53] and M3Grounder-Q (Qwen3-VL-8B) [50] with SAM [20] for segmentation. Training uses AdamW [29] with a learning rate 2×10^{-6} and 3% warmup followed by cosine decay. Gradients are clipped to a maximum norm of 1.0. Mixed precision is applied to the VLM; the mask decoder remains FP32 for stability. We use fixed loss weights, determined empirically. We set $\lambda_{lm}=1$ and $\lambda_{seg}=2$. For segmentation, the BCE term uses granularity-specific coefficients: $\lambda_{bce}^{(p)}=2.0$, $\lambda_{bce}^{(l)}=1.0$ and $\lambda_{bce}^{(b)}=0.5$ and similarly, the dice loss uses $\lambda_{dice}^{(p)}=1.0$, $\lambda_{dice}^{(l)}=0.5$ and $\lambda_{dice}^{(b)}=0.25$ for phrase, line, and block levels respectively. The auxiliary terms use $\lambda_{bleed}=0.5$ and $\lambda_{hier}=0.5$. All training experiments use 64H100 GPUs and DeepSpeed ZeRO-3 [39, 40] with an effective batch size 128.

Training Data: M3Grounder is trained on GroundingDocQA along with subsets of general DocVQA

datasets [12, 15, 32, 33, 47]. This joint training helps retain document QA ability while learning grounding. All finetuned models used for baseline comparison also use the same splits for fair evaluations.

Benchmarks: We evaluate across GroundingDocQA-Bench (ours), DOGR-Bench [65], MMDocBench [67], and BoundingDocs (test) [14]. Benchmark specific evaluation procedures are discussed in Sec. 5.1

Baselines: We compare M3Grounder against the following groups: Commercial VLMs [9, 19, 35], Open source VLMs [16, 30, 48–50, 53]. These are evaluated zero-shot using their best-performing prompts. We additionally Fine-Tune InternVL3.5 8B [53] and Qwen3-VL 8B [50] on the same training split. Since these models do not support hierarchical grounding or mask-based supervision, they are trained only at the phrase level bounding boxes. We also include Grounding Models designed specifically for grounding document QA. All evaluations report phrase-level grounding performance across benchmarks. For DOGR [65], scores are reported only on DOGRBench since the code [66] could not be reliably executed or adapted for fine-tuning.

5.1. Benchmarks and Metrics

We evaluate M3Grounder and all baselines across four document grounding benchmarks (discussed below). Since these benchmarks annotate evidence differently, some with pixel-level masks and others with bounding boxes, we convert model predictions to match the annotation type used by each benchmark.

Mask-Box Conversion: To enable fair comparison, we adopt a consistent conversion strategy: (i) if a model predicts *bounding boxes* and the benchmark provides *masks*, we rasterize each box into a binary mask (pixels inside the box set to 1, others to 0) and evaluate using mask IoU (ii) if a model predicts *masks* and the benchmark provides *bounding boxes*, we apply standard preprocessing to obtain the tightest axis-aligned box enclosing the predicted foreground before computing IoU in box space.

BoundingDocs-Test (1K docs, 3K QA, bounding box (cleaned subset)) [14]: Grounding is evaluated using $F1_g$ with $IoU > 0.5$ between predicted and ground truth boxes. Following prior work [25, 51, 64], we use G-Eval [27] as a unified semantic scoring metric for Answer Quality (AQ), complementing lexical and structural measures for balanced semantic-spatial evaluation.

DOGR-Bench (0.2K docs, 1.1K QA, bounding box) [65]: We follow the same evaluation procedure as BD-Bench for the boxes, i.e. IoU-based $F1_g$. For multi-span answers, we apply bipartite matching and G-Eval for AQ.

MMDocBench (2.4K docs, 4.3K QA, bounding box) [67]: We evaluate box grounding using Intersection

Model	Params	BD-Test		DOGR-Bench		MMDocBench			GroundingDocQA-Bench(Ours)				
		F1 _g	AQ	F1 _g	AQ	EM	F1 _{txt}	IoU	SS	MS	CS	F1 _g	AQ
GPT-4o [19]	–	2.3	85.7	2.9	83.8	71.9	73.2	2.5	5.2	0.8	4.4	3.4	80.1
GPT-5 [35]	–	5.4	89.3	3.6	87.0	74.3	76.6	9.3	5.3	2.1	4.8	4.5	83.5
Gemini-2.5-Pro [9]	–	<u>70.0</u>	87.9	<u>59.3</u>	<u>87.8</u>	<u>77.8</u>	<u>79.4</u>	<u>49.4</u>	<u>53.8</u>	<u>14.0</u>	<u>32.1</u>	<u>43.4</u>	83.1
Gemma-3 [48]	12B	0.5	70.1	1.77	67.5	58.4	59.9	1.6	1.3	1.8	2.1	1.5	65.6
Kimi VL [49]	16B	8.1	68.2	7.2	68.0	59.9	61.6	3.5	0.6	0.0	2.7	0.5	64.0
mPLUG-DocOwl2 [16]	9B	1.2	69.0	0.0	43.3	15.9	17.3	0.0	0.0	0.0	0.0	0.0	37.5
InternVL3.5 [53]	8B	41.5	83.8	12.5	80.9	65.7	67.4	15.5	9.3	2.6	6.3	7.6	80.0
Qwen3-VL [50]	8B	44.5	84.3	27.6	82.6	69.8	71.3	28.7	15.7	3.8	11.6	12.8	79.9
Kosmos 2.5 Chat [30]	1.3B	0.7	33.7	0.0	36.1	12.6	13.8	0.0	0.0	0.0	0.0	0.0	49.1
DOGR [65]	8B	-	-	66.4	78.3	-	-	-	-	-	-	-	-
InternVL3.5 ft. [53]	8B	58.7	84.6	27.6	79.9	69.6	71.7	37.6	58.7	34.5	33.7	52.7	80.9
Qwen3-VL ft. [50]	8B	62.3	82.6	35.8	80.7	71.6	72.8	43.4	68.1	37.7	38.3	60.6	81.9
M3Grounder-I (p)	8B	77.2	86.6	69.6	78.7	70.8	71.3	65.5	77.6	52.7	81.7	71.3	82.3
M3Grounder-Q (p)	8B	81.4	88.8	73.3	80.8	73.3	74.1	68.2	84.2	63.5	85.3	79.0	80.1

Table 1. **Quantitative results on document grounding benchmarks.** We evaluate all models at the phrase level across four benchmarks: BoundingDocs (test) (BD-Test) [14], DOGR-Bench [65], MMDocBench [67], and GroundingDocQA-Bench (ours). For BD-Test and DOGR-Bench, grounding is measured by F1_g (IoU>0.5) over bounding boxes, and answer quality (AQ) by G-Eval [27]. For MMDoc-Bench, we follow the official evaluation using Exact Match (EM) and word-level F1_{txt} for textual accuracy, and IoU for region alignment. For GroundingDocQA-Bench, SS, MS denote single and multi-span grounding F1 (IoU>0.5), CS measures F1 (IoU>0.5) on curved/skew answers, overall F1_g (IoU>0.5) aggregates across spans, and AQ uses G-Eval. We report results for Commercial VLMs, Open source VLMs, Grounding Models, and Fine-Tuned variants. Higher is better for all metrics. **Bold** denotes the highest among open-source models, and Underline denotes the highest among commercial models.

Model	Phrase	Line	Block
M3Grounder-I	71.3	74.9	82.5
M3Grounder-Q	79.0	81.37	87.5

Table 2. **Multi-granular grounding performance** measured by F1_g on GroundingDocQA-Bench. Note the consistent improvements as granularity increases from phrase to block level.

over Union (IoU) and answer correctness using character-level Exact Match (EM) and word-level F1_{txt}.

GroundingDocQA-Bench (Ours) (2.5K docs, 5K QA, mask at 3 levels): We report single-span (SS) F1 and multi-span (MS) F1 via bipartite matching [6], and F1 (CS) for curved and skewed subsets. F1_g is average over all questions. Answer Quality is evaluated using G-Eval.

6. Results

6.1. Overall Comparison

Both variants of M3Grounder achieve state-of-the-art grounding performance across all evaluated benchmarks (see bottom two rows of Tab. 1). Their gains are most pronounced on the curved-skewed text (CS) split of our GroundingDocQA-Bench, where M3Grounder substantially surpasses both commercial and open-source systems. This indicates that our segmentation-based grounding captures the geometry of irregular text regions more

faithfully than box-based localization approaches.

Gemini-2.5 Pro performs strongly in zero-shot conditions. Fine-tuning Qwen3-VL and InternVL3.5 on our GroundingDocQA yields notable improvements over base models, validating our data engine and our dataset’s effectiveness in improving document grounding capabilities for VLMs. The M3Grounder variants outperform even these fine-tuned baselines. This result highlights the benefit of dedicated segmentation head for structured grounding. Performance rises consistently from phrase to line to block level (see Tab. 2), with block-level grounding reaching the highest F1 scores (82.5 for M3Grounder-I and 87.5 for M3Grounder-Q), suggesting that the model leverages wider spatial context effectively.

On DOGR-Bench, M3Grounder-Q attains an overall grounding F1 of 73.3, exceeding DOGR by 6.9 points at the same backbone scale (8B). Similar trends appear across other benchmarks, confirming generalization across diverse document types. These results collectively demonstrate that segmentation-based grounding achieves more precise spatial alignment than bounding-box formulations.

6.2. General Document Understanding

Although M3Grounder is primarily optimized for answer grounding, it maintains strong performance on standard document QA benchmarks. As shown in Tab. 3, both

Model	Size	Doc VQA	Info VQA	Deep Form	KLC	WTQ	Tab Fact	Chart QA	Text VQA	Text Caps	Visual MRC
DocOwl-2 [16]	8B	80.7	46.4	66.8	37.5	36.5	78.2	70.0	66.7	131.8	217.4
InternVL3.5 [53]	8B	<u>92.3</u>	79.1	-	-	-	-	86.7	<u>78.2</u>	-	-
Qwen3-VL [50]	8B	96.1	83.1	-	-	-	-	-	-	-	-
DOGR [65]	8B	91.7	70.7	70.8	40.4	58.8	<u>84.5</u>	83.6	76.6	145.9	<u>332.5</u>
M3Grounder-I	8B	90.3	76.2	<u>71.9</u>	<u>41.6</u>	<u>59.7</u>	81.9	82.2	78.6	<u>147.2</u>	291.8
M3Grounder-Q	8B	92.1	<u>79.6</u>	72.8	42.8	60.4	85.3	<u>83.9</u>	78.1	148.1	334.2

Table 3. **Performance comparison of M3Grounder across 10 general document QA benchmarks:** We compare against other models of similar size. Bold indicates the highest, and underline indicates the second-highest scores.

Setting	BD-T	DR-B	MD-B	GR-B		
				Phrase	Line	Block
HIERARCHY						
w/o Hierarchy	74.8	58.2	54.7	71.3	-	-
SEGMENTATION						
1xShared MLP + SAM	64.2	61.7	53.5	63.6	69.3	77.6
LOSS COMPONENTS						
$\mathcal{L}_{lm} + \mathcal{L}_{seg}$	78.3	70.1	65.6	74.0	77.6	84.4
$\mathcal{L}_{lm} + \mathcal{L}_{seg} + \mathcal{L}_{hier}$	80.1	72.8	67.4	78.2	79.1	84.7
$\mathcal{L}_{lm} + \mathcal{L}_{seg} + \mathcal{L}_{bleed}$	79.6	72.4	67.7	77.3	78.7	86.5
PEFT						
LoRA	62.4	54.8	52.7	61.5	67.3	75.7
M3Grounder-Q	81.4	73.3	68.2	79.0	81.4	87.5

Table 4. **Ablation Summary.** Comparison across hierarchical supervision, loss design, and fine-tuning strategy. BD-T (BoundingDocs (test)), DR-B (DOGR-Bench), and MD-B(MMDocBench) are evaluated at phrase level. GR-B (GroundingDocQA-Bench) reports grounding performance at phrase, line, and block granularity. Higher is better.

M3Grounder-I and M3Grounder-Q achieve best or competitive scores across benchmarks [8, 31–33, 36, 43–47], demonstrating robust overall document understanding. This shows that enhancing localization precision need not come at the expense of reasoning ability, validating the balanced design of our model.

6.3. Ablation Studies

Removing hierarchical supervision and training only at the phrase level leads to a noticeable performance drop (see Tab. 4), particularly on GroundingDocQA-Bench (from 79.0 to 71.3). This highlights the importance of multi-granular supervision.

Replacing the three granularity-specific MLPs with SAM’s default (a single shared MLP that produces one unified prompt) also results in a performance drop across benchmarks (Tab. 4), motivating our design choice.

For the loss components, starting from the base objective $\mathcal{L}_{lm} + \mathcal{L}_{seg}$, adding the hierarchical enclosure loss consistently improves performance across datasets, confirming that enforcing nested containment stabilizes

multi-granular grounding. The bleed-suppression loss yields smaller yet consistent gains by mitigating region spillover into neighboring text.

Finally, we assess parameter-efficient fine-tuning (PEFT). Applying LoRA [17] only to attention projections substantially underperforms full fine-tuning across all settings, with the largest degradation observed at the phrase level—the most spatially fine-grained setting. This suggests that precise document grounding, especially at fine granularity, benefits from updating the full set of parameters rather than adapting a limited subset.

Limitations and Future Directions: Although mask-based grounding improves spatial fidelity, it increases computational cost compared to box-level methods. Also, our formulation currently operates at the single-page level. Extending hierarchical grounding to multi-page or long-context settings remains a promising open direction.

7. Conclusion

M3Grounder is a novel hybrid vision–language and segmentation model for grounded Document QA that unifies textual reasoning with precise spatial localization. By decoupling answer generation from spatial prediction, it achieves fine-grained grounding without compromising linguistic accuracy. This design yields strong performance not only on the introduced **GroundingDocQA-Bench** but also across other grounding benchmarks. Through mask-based hierarchical supervision and dedicated grounding heads, **M3Grounder** improves both localization precision and answer quality. Our experiments show that multi-granular mask-based grounding enhances interpretability, particularly on documents with curved or complex layouts. The accompanying **GroundingDocQA** and **GroundingDocQA-Bench** enable systematic evaluation of hierarchical grounding and support progress in spatially grounded document understanding.

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